

High Performance Programming [COMTEK3, & ESD3]
Lecture 1: Parallel Processing - Communication and Data Structures

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Instruction: *The exercise problems are designed to aid your understanding of the concepts covered during the lecture. They are intended to be discussed and solved in a group of 3 – 6 students. The most important part of the exercise is the **learning process**. It is strongly recommended that you avoid the usage of LLMs such as chatGPT and similar AI models to generate solutions **without trying to solve the problems first**.*

Exercise 1: Network Models

Given a network model with 8 processors connected in a hypercube topology:

- (a) Represent the topology as a graph, $G = (V, E)$, labelling each processor.
- (b) Calculate the maximum number of steps required for communication between any two processors.
- (c) Compare this with the communication steps required in a ring topology with the same number of processors.

Exercise 2: Communication Cost

Assume a message of size $M = 100$ needs to be transferred across a mesh topology with 16 processors (i.e., 4×4 grid). The following parameters are given:

- $t_s = 2$ units (message preparation time)
 - $t_w = 1$ unit (unit transfer time)
- (a) Derive the communication cost for one-to-all broadcast on this topology.
 - (b) Analyze how the cost changes when the topology is switched to a hypercube.

Exercise 3: Broadcast and Reduction

A distributed system with 16 processors is arranged in a 4×4 mesh topology. Each processor initially holds one element of a vector $V = [3, 6, 1, 5, 7, 4, 2, 9, 8, 3, 5, 1, 4, 2, 6, 7]$.

- (a) Design an algorithm to compute the sum of all elements using broadcast and reduction operations.
- (b) Explain the steps for the one-to-all broadcast phase to distribute data among all processors.
- (c) Calculate the communication cost for the all-to-one reduction phase to collect the result at the first processor.

Exercise 4: Scatter and Gather Operations

A single processor has an array $A = [3, 6, 1, 5, 7, 4, 2, 9, 8, 3, 5, 1, 4, 2, 6, 7]$ with 16 elements. The array needs to be distributed among 4 processors ($p = 4$) in chunks of 4 elements per processor using a scatter operation.

- (a) Describe the steps for the scatter operation in a ring topology.
- (b) Assume each processor computes the local sum of its elements after the scatter. Write the steps to collect these local sums back at the root processor using a gather operation.

- (c) Calculate the communication cost for both scatter and gather operations, assuming: $t_s = 2$, and $t_w = 1$.

Exercise 5: Data Structures in Parallel Systems

Match the following data structures with their advantages and challenges in high-performance computing:

- Arrays
 - Linked Lists
 - Hash Tables
 - Graphs
1. Efficient for dynamic memory usage but poor cache performance.
 2. Fast lookups but irregular data access patterns in parallel algorithms.
 3. Cache-friendly and ideal for vectorization but static in size.
 4. High memory costs for adjacency matrix representation but useful for modeling complex relationships.

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